When Better Return Forecasts Do Not Produce Better Portfolios

Group Project

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Machine Learning in Asset Pricing

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Abstract

This project studies the gap between statistical return predictability and economic portfolio performance. In empirical asset pricing, machine learning models are often evaluated according to their forecasting accuracy, for example through MSE or related prediction metrics. However, an accurate return forecast is not necessarily a useful investment signal. Once predicted returns are used inside a mean-variance optimisation problem, small forecasting errors can be amplified into large, concentrated and unstable portfolio weights.

To investigate this issue, we estimate stock-level Kernel Ridge Regression models using linear, polynomial and RBF kernels. The forecasts are then used as expected-return inputs in a regularised mean-variance portfolio. The resulting KRR-MVE strategies are compared with a simple equally weighted $1/N$ benchmark. The analysis is conducted under a static out-of-sample specification and is complemented by a rolling-window robustness check.

The results show that better or more flexible forecasting models do not automatically lead to better portfolios. Nonlinear kernels do not provide a robust improvement over the linear model. In particular, the $1/N$ benchmark is difficult to beat because it avoids expected-return estimation and covariance inversion.

1. Overview

This project studies the link between machine learning return forecasts and portfolio allocation. The main question is not only whether Kernel Ridge Regression (KRR) can forecast stock returns, but whether those forecasts create economic value once they are used in mean-variance optimisation.

For each stock i, we estimate an asset-specific prediction function:

$$R_{t+1,i} = g_i(\widetilde{z}_{t,i}) + \varepsilon_{t+1,i}, \tag{1}$$

where $R_{t+1,i}$ is the one-month-ahead return and $\widetilde{z}_{t,i}$ is the vector of standardized firm characteristics observed at time t.

We compare a linear kernel, used as the mandatory benchmark, with two nonlinear kernels: polynomial and Gaussian radial basis function (RBF). The predicted returns are then used as inputs in a regularized mean-variance efficient (MVE) portfolio. Portfolio performance is evaluated against the linear-kernel portfolio and the equal-weight benchmark.

2. Data and Preprocessing

We use the Gu–Kelly–Xiu stock-level dataset of monthly U.S. equities. The sample starts in January 1990. To keep the exercise computationally feasible, we restrict the analysis to 50 stocks and 10 firm characteristics.

The baseline universe is formed by selecting the 50 largest stocks at the initial date using market equity `mvel1`. We do not require firms to survive until the end of the sample. Hence, the reported results refer to the baseline specification without a survivor restriction.

The selected predictors are standard firm characteristics used in empirical asset pricing:

Variable	Category	Interpretation
`mvel1`	Size	Market equity
`bm`	Value	Book-to-market ratio
`mom12m`	Momentum	Twelve-month momentum
`chmom`	Momentum	Change in momentum
`operprof`	Profitability	Operating profitability
`invest`	Investment	Asset growth
`beta`	Risk	Market beta
`idiovol`	Risk	Idiosyncratic volatility
`ill`	Liquidity	Illiquidity
`turn`	Trading activity	Share turnover

Table 1: Selected firm characteristics

The target variable is the next-month return. Since the version of the dataset used in this project does not contain a separate realized return variable for each asset, we construct a return proxy using `mom1m`. Specifically, `mom1m` is shifted forward within each stock:

$$R^{proxy}_{t+1,i} = \texttt{mom1m}_{t+1,i}. \tag{2}$$

Thus, characteristics observed at month t are used to predict a return proxy realized at month $t+1$. After constructing the target, `mom1m` is removed from the predictors.

Missing characteristics are filled with the monthly cross-sectional median. Then, each characteristic is standardized cross-sectionally within each month:

$$\widetilde{Z}_{t,i,k} = \frac{Z_{t,i,k} - \overline{Z}_{t,k}}{s_{t,k}}, \tag{3}$$

where $\overline{Z}_{t,k}$ and $s_{t,k}$ are the cross-sectional mean and standard deviation of characteristic k at month t. This scaling is required because kernel methods are sensitive to the relative magnitude of their inputs.

Finally, the panel is split chronologically into training, validation and test samples.

Sample	**Share**	**Purpose**
Training	60%	Estimate KRR models and covariance matrix
Validation	20%	Select λ_i, kernel parameters and MVE γ
Test	20%	Final out-of-sample evaluation

Table 2: Chronological sample split

The test set is used only once, after all hyperparameters have been fixed.

3. Methodology

This section describes the forecasting model, the kernel specifications, the hyperparameter selection procedure and the construction of the mean-variance portfolios.

3.1 Kernel Ridge Regression

For each asset i, we estimate the asset-specific model

$$R_{t+1,i} = g_i(\widetilde{z}_{t,i}) + \varepsilon_{t+1,i}, \tag{4}$$

where $\widetilde{z}_{t,i} \in \mathbb{R}^K$ is the vector of standardized characteristics observed at time t.

Given a kernel function $\kappa(\cdot,\cdot)$, the asset-specific kernel matrix is

$$K^{(i)}_{t,s} = \kappa\left(\widetilde{z}_{t,i}, \widetilde{z}_{s,i}\right), \qquad t,s \in \mathcal{T}_{train}. \tag{5}$$

The KRR coefficients are computed as

$$\widehat{\alpha}^{(i)} = \left(K^{(i)} + \lambda_i I\right)^{-1} R^{(i)}, \tag{6}$$

where $R^{(i)}$ is the vector of training returns and $\lambda_i > 0$ is the asset-specific ridge penalty.

For an out-of-sample date t^*, the predicted return is

$$\widehat{R}_{t^*+1,i} = \sum_{t \in \mathcal{T}_{train}} \widehat{\alpha}^{(i)}_t \kappa\left(\widetilde{z}_{t^*,i}, \widetilde{z}_{t,i}\right). \tag{7}$$

3.2 Kernel functions

We compare three kernel specifications.

The first specification is the linear kernel:

$$\kappa_{lin}(z, z') = z^\top z'. \tag{8}$$

This kernel is the benchmark specification and captures only linear relations between firm characteristics and future returns.

The second specification is the polynomial kernel:

$$\kappa_{poly}(z, z') = \left(c + z^\top z'\right)^d. \tag{9}$$

This kernel allows for interaction effects among characteristics through the polynomial degree d.

The third specification is the Gaussian radial basis function kernel:

$$\kappa_{rbf}(z, z') = \exp\left(-\gamma_{rbf} \left\|z - z'\right\|^2\right). \tag{10}$$

This kernel captures nonlinear similarity patterns in the characteristic space, where γ_{rbf} controls how local the comparison between observations is.

3.3 Hyperparameter selection

Hyperparameters are selected on the validation set by minimizing

$$MSE_{val}^{(i)} = \frac{1}{|\mathcal{T}_{val}|} \sum_{t \in \mathcal{T}_{val}} \left(R_{t+1,i} - \widehat{R}_{t+1,i}\right)^2. \tag{11}$$

The ridge penalty λ_i is selected separately for each asset. Kernel-specific parameters are selected by validation MSE, while the test set is reserved for the final out-of-sample evaluation.

3.4 Mean-Variance Portfolio Construction

At each test date t^*, predicted returns are collected in the vector $\widehat{R}_{t^*+1} \in \mathbb{R}^N$ and used as expected-return inputs in a regularized MVE allocation:

$$w_{t^*} = \left(\widehat{\Sigma} + \gamma I_N\right)^{-1} \widehat{R}_{t^*+1}, \tag{12}$$

where $\widehat{\Sigma}$ is the covariance matrix estimated from past returns and $\gamma > 0$ stabilizes the matrix inversion.

Weights are rescaled to unit gross exposure,

$$\sum_{i=1}^{N} |w_{t^*,i}| = 1, \tag{13}$$

and the realized portfolio return is

$$R_{t^*+1}^{MVE} = w_{t^*}^\top R_{t^*+1}. \tag{14}$$

4. Empirical Results

The main empirical results are based on the static out-of-sample specification. The KRR models are estimated once on the development sample, defined as the union of the training and validation samples. The covariance matrix is also estimated once on the development sample. Portfolio weights are then recomputed monthly during the test period because predicted returns change over time.

4.1 Forecasting performance

Table 3 reports the out-of-sample forecast accuracy for the static specification. The linear kernel obtains the lowest test MSE, RMSE and MAE. The RBF kernel is close, while the polynomial kernel has the weakest forecasting performance in this specification.

Table 3: Static out-of-sample forecast accuracy

Model	Test MSE	Test RMSE	Test MAE
Linear	0.005134	0.071649	0.050385
Polynomial	0.005261	0.072532	0.051131
RBF	0.005220	0.072252	0.051057

The differences in forecast errors are small. This is important because the portfolio optimisation step can still transform small differences in expected returns into large differences in weights.

4.2 Model selection across assets

Table 4 summarizes the validation-stage model selection. The RBF kernel has the lowest average validation MSE and is selected for most stocks. However, this validation advantage does not automatically imply superior portfolio performance in the test period.

Table 4: Validation model selection across assets

Kernel	Mean Validation MSE	Stocks Selected
Linear	0.008305	0
Polynomial	0.008118	2
RBF	0.007817	31

The portfolio regularization parameters selected on the validation sample are $\gamma_{\text{MVE}} = 10^4$ for the linear and polynomial kernels, and $\gamma_{\text{MVE}} = 0.1$ for the RBF kernel.

4.3 Portfolio performance

Table 5 reports the static out-of-sample portfolio performance. The $1/N$ benchmark has the highest annualized return, Sharpe ratio and final wealth. Among the KRR-MVE portfolios, the linear kernel is the strongest, but it still does not outperform the equally weighted benchmark.

Table 5: Static out-of-sample portfolio performance

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max DD	Final Wealth
$1/N$	0.1016	0.1430	0.7106	-0.2376	1.7945
Linear KRR-MVE	0.0709	0.1176	0.6031	-0.1686	1.5070
Polynomial KRR-MVE	0.0184	0.1190	0.1549	-0.1870	1.0762
RBF KRR-MVE	0.0393	0.1035	0.3797	-0.1876	1.2434

The static results show that lower forecast error is not sufficient to beat $1/N$. The KRR-MVE portfolios reduce volatility relative to the benchmark, but the reduction in risk is not enough to compensate for their lower realized returns.

4.4 Cumulative wealth

Figure 1 plots cumulative wealth for the static specification.

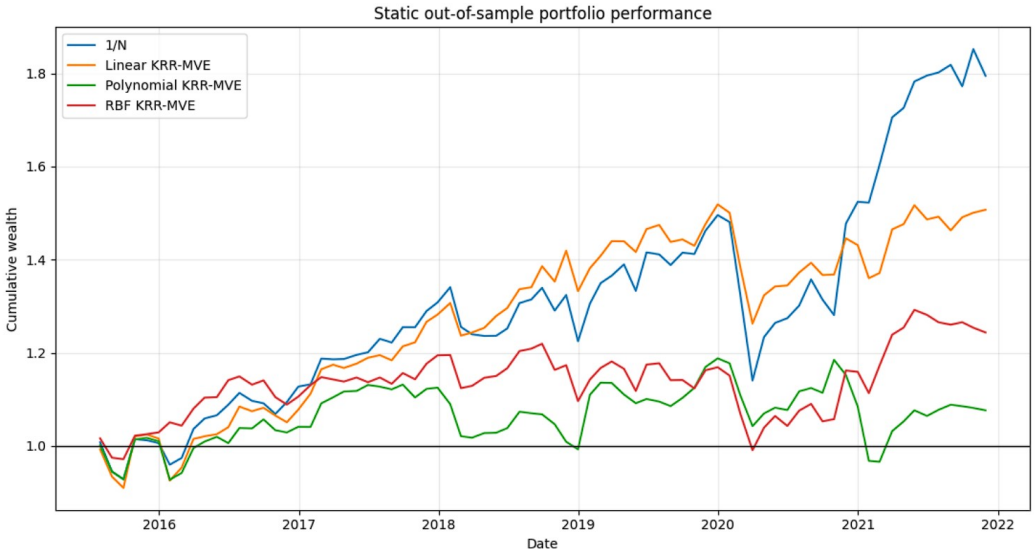


Figure 1: Static cumulative wealth

The graph is consistent with Table 5. The $1/N$ benchmark remains the strongest strategy over the test period, while the KRR-MVE portfolios produce lower terminal wealth.

4.5 Weight diagnostics

Table 6 reports the static weight diagnostics. Gross exposure is equal to one for all strategies, confirming that the normalization is working correctly:

$$\sum_i |w_{i,t}| = 1.$$

Table 6: Static weight diagnostics

Strategy	Gross Exp.	Net Exp.	Conc.	Max Abs. W.	Turnover
$1/N$	1.0000	1.0000	0.0352	0.0352	0.0013
Linear KRR-MVE	1.0000	0.6153	0.0897	0.1741	0.1426
Polynomial KRR-MVE	1.0000	0.4976	0.1248	0.2442	0.1909
RBF KRR-MVE	1.0000	0.5906	0.0619	0.1441	0.1436

The KRR-MVE portfolios are more concentrated and have higher turnover than $1/N$. This suggests that the optimisation step creates less stable allocations even when leverage is controlled.

5. Robustness and Sensitivity Analysis

This section reports the rolling-window robustness check and the two sensitivity analyses.

5.1 Rolling-window robustness

In the rolling-window specification, the KRR models are refitted every test month using the previous 120 months of data. The covariance matrix is re-estimated on the same rolling window. Hyperparameters remain fixed at the validation-selected values. The rolling-window test uses 77 months and 2193 stock-month observations, the same test coverage as the static specification.

Table 7: Rolling-window forecast accuracy

Model	Test MSE	Test RMSE	Test MAE
Linear	0.005122	0.071571	0.050234
Polynomial	0.005187	0.072020	0.050625
RBF	0.005421	0.073626	0.051478

The rolling-window forecast results again do not support a clear advantage for nonlinear kernels. The linear model has the lowest forecast error.

Table 8: Rolling-window portfolio performance

Strategy	Ann. Return	Ann. Vol.	Sharpe	Max DD	Final Wealth
$1/N$	0.1016	0.1430	0.7106	-0.2376	1.7945
Linear KRR-MVE	0.0807	0.1146	0.7043	-0.1221	1.6080
Polynomial KRR-MVE	0.0538	0.1247	0.4314	-0.1669	1.3429
RBF KRR-MVE	0.0500	0.1138	0.4396	-0.1619	1.3239

The rolling-window results are slightly stronger for the KRR-MVE portfolios, especially for the linear kernel. However, the $1/N$ benchmark still has the highest final wealth and the highest Sharpe ratio.

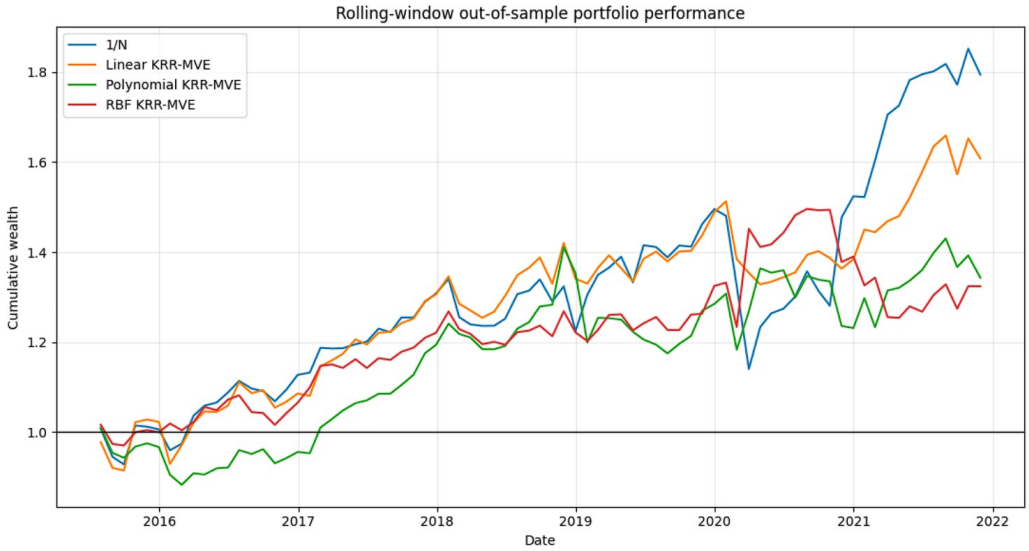


Figure 2: Rolling-window cumulative wealth

Table 9: Rolling-window weight diagnostics

Strategy	Gross Exp.	Net Exp.	Conc.	Max Abs. W.	Turnover
$1/N$	1.0000	1.0000	0.0352	0.0352	0.0013
Linear KRR-MVE	1.0000	0.8095	0.1033	0.2088	0.2093
Polynomial KRR-MVE	1.0000	0.6861	0.1106	0.2289	0.2921
RBF KRR-MVE	1.0000	0.5752	0.0818	0.1830	0.3587

The rolling-window portfolios remain more concentrated and more active than the benchmark. Turnover increases relative to the static specification, especially for the RBF and polynomial portfolios.

5.2 Sensitivity to KRR regularization

The first sensitivity analysis studies the KRR regularization parameter λ, it checks how validation MSE changes across different values of the KRR penalty.

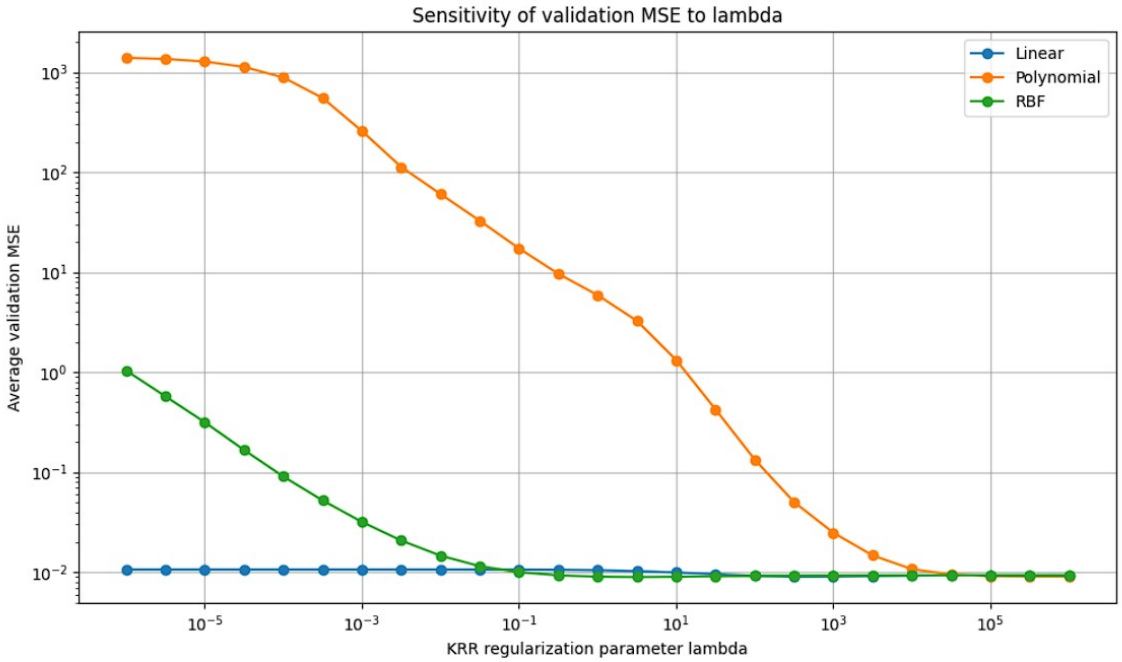


Figure 3: Sensitivity of validation MSE to KRR regularization

Since the vertical axis is in logarithmic scale, differences should be interpreted in orders of magnitude rather than in absolute units. The linear kernel is stable across the grid, with validation MSE remaining close to 10^{-2}. In contrast, the polynomial kernel is highly unstable for small values of λ, with validation errors several orders of magnitude larger, suggesting overfitting or numerical instability. The RBF kernel shows an intermediate pattern and also benefits from stronger regularization.

Overall, the figure indicates that nonlinear kernels are more sensitive to the choice of λ.

5.3 Sensitivity to MVE regularization

The second sensitivity analysis studies the MVE regularization parameter γ_{MVE}. Forecasts are kept fixed, while the RBF KRR-MVE portfolio is recomputed for different values of γ_{MVE}.

The RBF portfolio is sensitive to γ_{MVE}. Higher values improve performance until the results reach a plateau. This suggests that stronger regularization can stabilize the MVE allocation. The table is interpreted only as an ex-post sensitivity check, not as a re-selection of the final test-period parameter.

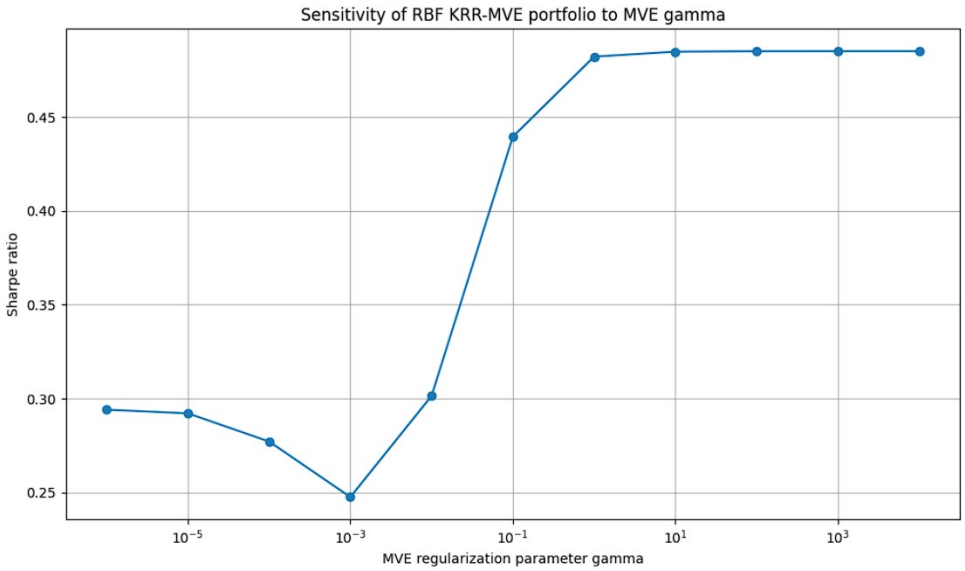


Figure 4: Sensitivity of RBF KRR-MVE Sharpe ratio to MVE γ

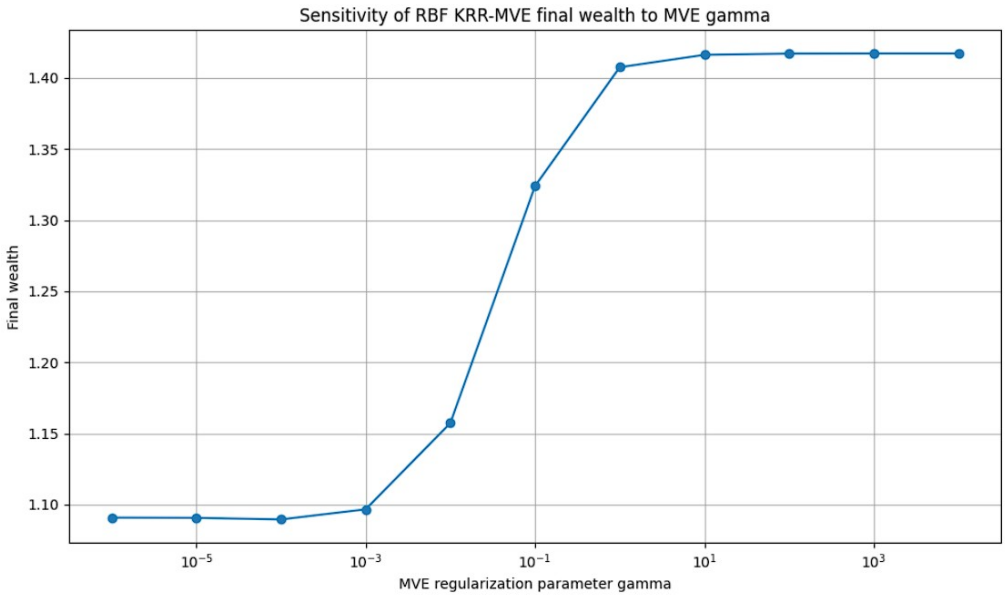


Figure 5: Sensitivity of RBF KRR-MVE final wealth to MVE γ

6. Asset Pricing Implications and Conclusion

6.1 Do nonlinear kernels improve performance?

The results do not show a robust improvement from nonlinear kernels. The RBF kernel performs best in validation model selection, but the linear kernel has the best test forecast accuracy and the strongest KRR-MVE portfolio performance. The polynomial kernel is consistently weaker.

6.2 Sensitivity of the results

The results are sensitive to the portfolio regularization parameter. The MVE γ sensitivity shows that stronger regularization can improve the RBF portfolio. However, even after rolling refitting and sensitivity checks, the model-based portfolios do not clearly dominate the $1/N$ benchmark.

6.3 Economic interpretation

The main economic point is that forecast accuracy and portfolio performance are not the same object. Mean-variance optimisation transforms predicted returns into weights. When expected returns are noisy, small forecast errors can generate concentrated and high-turnover portfolios.

6.4 Conclusion

The evidence show that the linear KRR-MVE strategy is the best model-based portfolio, in KRR framework, especially in the rolling-window specification, but it still does not outperform 1/N in final wealth or Sharpe ratio. Overall, the results support the central conclusion: better or more flexible return forecasts do not necessarily produce better portfolios. In machine-learning asset pricing, economic value depends not only on prediction accuracy, but also on the stability of the optimisation step that converts forecasts into portfolio weights

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